Adaptive Queue Length + Fuzzy Logic Congestion Control in Clustered WSN

This document describes the final design for congestion control in a clustered Wireless Sensor Network (WSN). The solution combines **adaptive queue length adjustment** at Cluster Heads (CHs) and **independent fuzzy logic–based congestion detection** using Active Queue Management.

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1. Overview of the Proposed Design

Our system contains two independent components:

A. Adaptive Queue Length Mechanism (AQLM)

- Queue length at each CH is *not fixed*.
- It is dynamically adjusted based on:
- **Packet Loss Ratio (PLR)**
- **Downstream/Neighbour Cluster Head Capacity**
- AQLM runs **only when PLR exceeds a threshold**.

B. Fuzzy Logic-Based Congestion Detection (FIS + AQM)

- Runs periodically, independent of AQLM.
- Inputs include:
- Average Queue Length (AQL)
- Packet Loss Ratio (PLR)
- Arrival Rate
- Residual Energy
- Packet Priority
- Current Queue Capacity
- Outputs:
- Drop Probability
- Throttling/Rate Adjustment
- Congestion Notification Flag

The two systems **do not control each other internally**, but AQLM affects the queue state which the fuzzy engine reads as an input.

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2. Workflow of the System

Step 1: Measure metrics periodically

- PLR updated via EWMA

- AQL calculated from queue occupancy
- Neighbouring CH capacity obtained via control adverts

Step 2: If PLR > threshold, adapt queue length

- Compute desired queue length using PLR
- Scale desired queue using downstream capacity
- Apply smoothing/hysteresis to avoid oscillation
- Update the queue size

Step 3: Fuzzy Logic runs independently

- Reads current queue size and other inputs
- Produces AQM actions
- Handles congestion detection + response

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3. Control/Advert Packet Format (Lightweight)

Neighbour CHs periodically advertise their capabilities using a small, piggybacked message.

Advert Packet Structure

```
{
  CH_ID:          uint8,
  service_rate:   float (EWMA of pkts/sec),
  avail_buffer:   uint8 (free queue slots),
  energy_level:   uint8 (optional),
  timestamp:      uint32
}
```

- Sent every **T_adv = 2–5 seconds** - Missing advert for **T_timeout** ⇒ assume low capacity

Use in algorithm

- `service_rate` used to compute downstream capacity
- `avail_buffer` used as optional buffer-awareness factor

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4. Adaptive Queue Length Algorithm (AQLM)

Parameters

- `Q_min`, `Q_max` – allowed queue limits
- `PLR_th` – threshold at which adaptation begins

- `C_ref` – reference send rate of the CH
- `β` – smoothing factor
- `ΔQ_min` – minimum effective change
- `step_max` – max queue adjustment per update

Algorithm

```
Every T_meas seconds:
  Measure PLR_sample
  PLR = α*PLR_sample + (1−α)*PLR

  If PLR > PLR_th and time_since_last_update >= T_hold:
      Q_des_base = Q_min + ((PLR−PLR_th)/(1−PLR_th)) * (Q_max − Q_min)
      C_down = compute_downstream_capacity()   # EWMA over neighbour adverts
      γ = clamp(C_down / C_ref, 0, 1)

      Q_des = clamp(Q_min + γ*(Q_des_base − Q_min), Q_min, Q_max)

      Q_proposed = β*Q_des + (1−β)*Q_current

      If |Q_proposed − Q_current| >= ΔQ_min:
          change = sign(Q_proposed − Q_current) * min(step_max, |Q_proposed −
Q_current|)
          Q_current = Q_current + change
          record_update_time()
```

Notes

- Avoids sudden jumps in queue length
- Prevents cascading congestion due to neighbour overload
- Only triggers under sustained high PLR

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5. Fuzzy Inference System (FIS) — Runs Independently

Inputs

- Average Queue Length (AQL)
- Packet Loss Ratio (PLR)
- Arrival Rate
- Packet Priority
- Residual Energy
- **Current Queue Capacity** (output of AQLM)

Outputs

- Drop Probability
- Transmission Rate Adjustment
- Congestion Notification (ECN flag)

Operation

- Runs every **T_FIS** seconds
- Reads the *current* queue size, but does not influence its calculation

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6. Neighbour Capacity Calculation

```
For each neighbour i:
  C_i = EWMA(service_rate_i)

C_down = min_i(C_i)   # conservative

Optionally:
C_down = weighted_avg(C_i, weights=link_quality)
```

- If no advert received: treat neighbour as low-capacity
- Ensures stability and avoids cascade failures

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7. Numerical Example (Short)

- `Q_min = 10`, `Q_max = 100`
- PLR threshold = 0.10, measured PLR = 0.28
- Compute base: `Q_des_base = 28`
- Downstream capacity factor: `γ = 0.5`
- Final: `Q_des = 19`
- After smoothing → `Q_current` only increases gradually (avoids spikes)

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8. Advantages of the Design

- **Independent, modular controls** (robust and easy to tune)
- **Stable queue adaptation** avoids oscillation
- **Neighbour-aware** adjustment prevents congestion cascades
- FIS focuses purely on **detecting and responding** to congestion
- Adaptation triggers only when needed (high PLR)

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9. Implementation Notes

- EWMA parameters should be tuned depending on simulation
- Use different timing for AQLM and FIS to maintain independence
- Limit control packet size to maintain energy efficiency

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10. Conclusion

This final design provides a novel, hybrid congestion control mechanism for clustered WSNs by: - Dynamically adapting queue length based on PLR and neighbour capacity - Detecting congestion using a separate fuzzy logic AQM engine - Ensuring stability, energy efficiency, and non-cascading operation

It is simple, modular, and directly implementable in simulations such as NS-2/NS-3 or MATLAB.